

Agent-Analytics Framework for Physics-Informed Crease Modeling in Multilayer Flexible Cover Stacks

Kyongtae Park

AI TF of Mobile Business Samsung Display Co., Ltd., E-mail: ktman.park@samsung.com

Abstract

Crease formation in foldable displays is a critical quality issue governed by the complex interplay of viscoelastic relaxation, structural asymmetry, and bending mechanics. As folding radii decrease, the residual stress manifests as a visible surface deformation, degrading both perceived image quality and mechanical reliability. While Finite Element Analysis (FEA) provides mechanistic insights, it is computationally prohibitive for exploring high-dimensional design spaces and depends heavily on complex constitutive models. Conversely, data-driven Machine Learning (ML) lacks physical interpretability. This paper introduces a Physics-Informed Agent-Analytics Framework that combines human domain expertise, rigorous physics-based feature engineering, and ensemble Explainable AI (XAI). Moving beyond a passive assistant role, the AI agent served as a core mediator for knowledge translation, converting qualitative engineering hypotheses into rigorous mathematical descriptors using the Parallel Axis Theorem (PAT) to accurately account for the neutral axis shift and the "lever-arm" effect of effective stiffness. To ensure practical validity, we constructed a dataset of 50 distinct stack configurations, which were not merely simulated but physically fabricated and experimentally tested. To ensure robustness, we employed an ensemble of 11 ML algorithms (including Random Forest, Gradient Boosting, and SVM) and computed SHAP values with confidence intervals. Our analysis reveals that crease severity is primarily driven by Viscoelastic Accumulation and Stiffness Symmetry, followed by Elastic Strain Energy. Counter-intuitively, lower stored elastic energy correlates with reduced crease, suggesting that stress relief in high-modulus layers is critical. Finally, a comparative case study demonstrates that a "Balanced Design" strategy—optimizing structural symmetry alongside material properties—yields superior and more practical crease mitigation than extreme material reduction alone.

Author Keywords

Physics-Informed Machine Learning, Agent-Analytics, Crease Modeling, SHAP Explainability, Stiffness Symmetry Index

1. Introduction

As the form factor of mobile devices evolves toward foldable and rollable designs, the mechanical durability and optical quality of the multilayer cover stack have become paramount. A persistent challenge in this domain is the "crease"—a residual surface deformation at the folding area caused by the interplay of long-term viscoelastic creep, plastic deformation of adhesive layers, and localized stress concentration [1], [2]. This phenomenon is particularly exacerbated in heterogeneous stacks comprising materials with vastly different moduli, such as Ultra-Thin Glass (UTG), optical polymers, and Pressure Sensitive Adhesives

(PSAs). Traditionally, Finite Element Analysis (FEA) has been the gold standard for predicting such deformations. However, accurate FEA requires precise nonlinear visco-hyperelastic constitutive models, which are often unavailable for novel developmental materials. Furthermore, simulating thousands of stacking combinations to explore the high-dimensional design space involves significant computational cost and time [3]. On the other hand, conventional Machine Learning (ML) can predict outcomes rapidly but typically treats the multilayer stack as a "black box." It often relies on raw geometric parameters (e.g., individual layer thicknesses) without capturing the underlying mechanics, thus failing to explain why a specific stack performs well or offering actionable guidance for engineering design.

To bridge this gap, we propose a Physics-Informed Agent-Analytics Framework (Fig. 1). This novel approach leverages a collaborative workflow where human domain experts and Large Language Model (LLM) agents co-design mechanical descriptors. Recent studies have demonstrated the potential of agents in autonomous reasoning and tool integration [4], yet applying them to complex mechanics requires domain-specific grounding. The core innovation lies in the Agent's role as a "Knowledge Translator": it effectively bridged the gap between abstract engineering intuition and formal mathematical modeling. Furthermore, unlike studies relying solely on virtual simulations, our framework is validated against a dataset of 50 physically fabricated multilayer stacks, ensuring that the insights are grounded in real-world phenomenology. Through this collaboration, we moved beyond simple geometric approximations to apply the Parallel Axis Theorem (PAT), defining high-level features such as Stiffness Symmetry Index and Elastic Strain Energy. These physics-informed descriptors are then analyzed using a robust ensemble ML model to provide statistically reliable and mechanically interpretable design guidelines.

2. Methodology

2.1. The Agent-Analytics Workflow: Bridging Intuition and Math

The core innovation of this framework is the utilization of an LLM agent not merely as a tool, but as a pivotal "Knowledge Mediator" to bridge the gap between qualitative engineering intuition and quantitative ML features. In traditional workflows, domain experts often struggle to translate abstract concepts (e.g., "The stack needs to be balanced") into executable code for high-dimensional datasets. Conversely, data scientists often lack the mechanical context to select appropriate features. Our Agent-Analytics workflow addresses this "Translation Gap" through three specific roles:

Knowledge Formalization: The human expert provided high-level mechanical hypotheses (e.g., "Crease is driven by the lever-

arm effect of outer layers"). The Agent successfully executed this translation, specifically proposing the application of the Parallel Axis Theorem (PAT) to quantify the effective stiffness relative to the neutral axis.

Automated Logic Synthesis: The Agent generated robust Python algorithms to automate the calculation of complex features—such as the Stiffness Symmetry Index and Weighted Viscoelastic Accumulation—while implementing safety checks for missing material properties (e.g., handling layers with zero thickness or missing relaxation data).

Iterative Refinement: Through a feedback loop, the Agent and human refined the descriptors. For instance, when initial simple stiffness models ($E t^3$) showed low correlation with crease data, the Agent suggested incorporating the Steiner term ($t d^2$) to account for the laminate structure, which significantly improved the model's explanatory power. This collaborative process transforms the LLM from a passive tool into an active participant in the feature engineering process, accelerating the discovery of physics-informed descriptors.

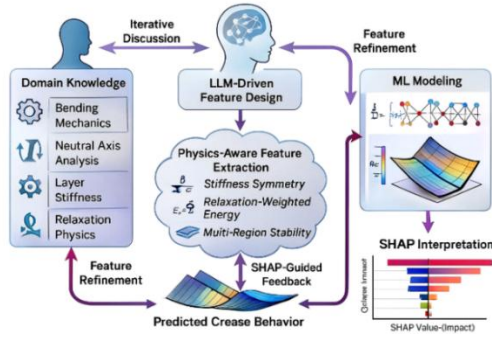


Fig. 1. Agent-Analytics workflow integrating domain reasoning, LLM descriptor formulation, physics-aware feature extraction, ML modeling, and SHAP interpretation.

2.2. Physics-Informed Feature Formulation

A critical limitation of previous data-driven studies was the oversimplification of layer stiffness (often approximated as $\propto t^3$). In this work, we apply the Parallel Axis Theorem (PAT) to rigorously define the effective stiffness. This method accounts for the distance of each layer from the neutral axis (NA), which significantly amplifies the contribution of outer layers to the total bending moment.

Neutral Axis and Effective Stiffness:

First, the global neutral axis position (z_{NA}) is calculated as the modulus-weighted centroid of the stack:

$$z_{NA} = \frac{\sum_{i=1}^N E_i t_i z_i}{\sum_{i=1}^N E_i t_i} \quad (1)$$

where E_i , t_i and z_i denote the elastic modulus, thickness, and centroid coordinate of the i -th layer, respectively.

Using z_{NA} , the effective bending stiffness (D_i) of each individual layer is derived as:

$$D_i = E_i \left(\frac{t_i^3}{12} + t_i (z_i - z_{NA})^2 \right) \quad (2)$$

The term $t_i (z_i - z_{NA})^2$ represents the Steiner term. For thin layers positioned far from the NA (e.g., the cover window or

backplate), this term dominates the local stiffness ($t^3/12$), creating a "lever-arm" effect that is crucial for accurate mechanical modeling (Fig. 2).

Key Physics-Informed Descriptors:

Based on the computed D_i , we formulated four critical descriptors to capture the dominant mechanisms:

1) Stiffness Symmetry Index (S_{sym}): This index quantifies the structural balance of the stack.

$$S_{sym} = \frac{\min(\sum_{top} D_i, \sum_{bot} D_i)}{\max(\sum_{top} D_i, \sum_{bot} D_i)} \quad (3)$$

A value approaching 1.0 indicates a mechanically symmetric structure. In a symmetric bend, the tensile and compressive stresses largely cancel out at the neutral plane, minimizing the net residual moment that drives crease formation.

2) Elastic Strain Energy Capacity ($U_{elastic}$): This represents the potential energy stored in purely elastic layers (where $E > 10 \text{ GPa}$ and relaxation $R \approx 0$) during bending.

$$U_{elastic} = \sum_{i \in \text{Elastic}} D_i \quad (4)$$

While often associated with recovery force, high stored energy also implies high internal stress, which can force adjacent viscoelastic layers into plastic flow. Since all configurations are evaluated at a constant folding radius (i.e., identical curvature κ), the bending strain energy $U = \frac{1}{2} \kappa^2 \sum D_i$ is directly proportional to $\sum D_i$, which we thus adopt as a curvature-independent proxy.

3) Viscoelastic Accumulation (V_{acc}): A weighted metric for creep susceptibility, defined as $\sum R_i (t_i/E_i)$. This formulation highlights that creep is most severe in thick, low-modulus layers with high relaxation ratios (e.g., soft PSAs). The ratio t_i/E_i represents the compliance-thickness product of each layer, quantifying its deformability per unit width. Weighting by the relaxation ratio R_i selectively amplifies layers prone to time-dependent flow, making V_{acc} an effective proxy for the total creep susceptibility of the stack.

4) Total Geometric Thickness (T_{total}): The baseline constraint defined as the simple sum of all layer thicknesses.

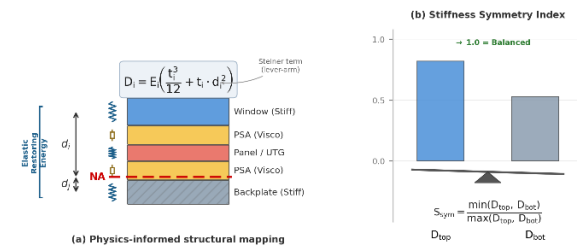


Fig. 2. Physics-informed structural mapping. (Left) Elastic Restoring Energy is accumulated in high-modulus layers (depicted as blue springs), amplified by their distance from the Neutral Axis (NA) due to the Parallel Axis Theorem. (Right) The Stiffness Symmetry Index quantifies the bending balance between the top (D_{top}) and bottom (D_{bot}) stacks.

2.3. Robust Ensemble Modeling Strategy

To ensure the reliability of our analysis and eliminate model-

specific biases, we employed an ensemble of 11 diverse regression algorithms, including Random Forest, Gradient Boosting (GBM), Support Vector Machines (SVM), CatBoost, and Gaussian Processes.

Instead of selecting a single "best" model, we aggregated the predictions and explanations. We computed SHAP (SHapley Additive exPlanations) values across all models to generate importance scores with confidence intervals (error bars). This rigorous approach confirms that the identified key factors are universal physical drivers, not artifacts of a specific algorithm.

3. Results and Discussion

3.1. Statistical Validation of Descriptors

To validate the discriminative power of the proposed descriptors, we categorized the dataset into "Good" (lowest 25% crease) and "Bad" (highest 25% crease) groups (Fig. 3). The boxplot analysis reveals clear separation between the groups. Specifically, the "Good" group consistently exhibits significantly lower V_{acc} , higher S_{sym} , and lower $U_{elastic}$. This strong statistical distinction confirms that our physics-informed features effectively capture the mechanical boundaries between superior and inferior stack designs.

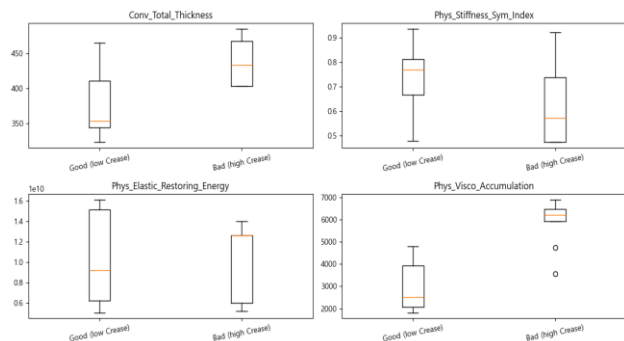


Fig. 3. Statistical validation of descriptors. Boxplot comparison between Good (low crease) and Bad (high crease) groups. The "Good" group shows distinct distributions with higher symmetry and lower viscoelastic accumulation, demonstrating the discriminative power of the proposed features.

3.2. Global Interpretability via Robust Ensemble

To ensure that the mechanical insights are physically intrinsic and not artifacts of a specific algorithm, we utilized an ensemble of 11 heterogeneous ML models, including Random Forest, Gradient Boosting, SVM, and CatBoost. The final importance scores were derived by computing the weighted average of SHAP values across all algorithms based on their predictive accuracy.

Figure 4 presents this consolidated SHAP importance, representing the consensus of the multi-model system. The statistical robustness of these predictions is visually confirmed in the detailed case study (Fig. 5), where error bars explicitly indicate the low variance among the ensemble members, validating the stability of our findings.

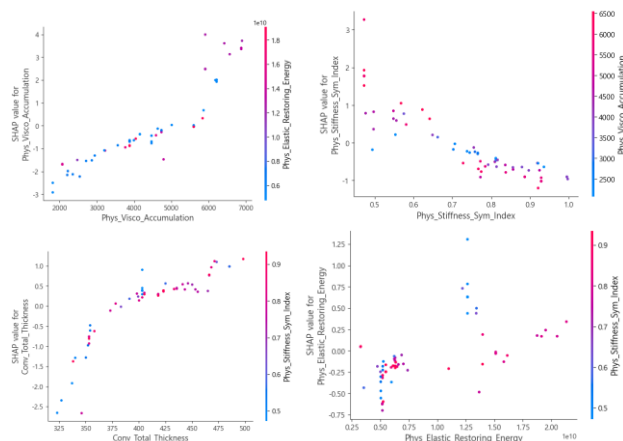


Fig. 4. Partial SHAP importance ranking. The values represent the weighted average from the 11-model ensemble, illustrating the consensus on the dominant drivers: Viscoelastic Accumulation and Stiffness Symmetry.

As shown in the summary plot (Fig. 4), the hierarchy of crease drivers is consistent:

Viscoelastic Accumulation (V_{acc}): Ranked as the paramount factor with a strictly monotonic trend. Minimizing the volume and relaxation ratio of soft layers is the primary defense against crease, as it directly reduces the source of time-dependent deformation.

Stiffness Symmetry Index (S_{sym}): Identified as the second most critical factor. The analysis shows that higher symmetry significantly reduces predicted crease depth. This implies that even with identical materials, optimizing the stacking order to balance the stiffness above and below the NA can yield substantial performance gains.

Elastic Strain Energy ($U_{elastic}$): Interestingly, our analysis reveals that lower elastic energy correlates with better crease performance. While high modulus is often associated with a strong "snap-back" force, excessive stiffness in the wrong position accumulates high internal stress. This stress can exceed the yield limit of adjacent soft layers, inducing permanent set. Thus, a "Soft-Elastic" design strategy—minimizing stored energy while maintaining necessary rigidity—is preferable to a "Hard-Elastic" one.

From Material Dependence to Geometric Compensation Historically, crease mitigation has heavily relied on improving the physicochemical properties of PSAs (e.g., stress relaxation or recovery rates). However, material development is inherently time-consuming and prone to reaching asymptotic physicochemical limits. Our findings regarding (S_{sym}) propose a fundamental paradigm shift from a "material-dependent" approach to "geometric compensation." The analysis confirms that by optimizing structural symmetry, the shear load transferred to the PSA layers can be intrinsically reduced. This implies that structural balance can effectively compensate for material limitations. Consequently, this "Geometric Compensation" strategy empowers engineers to bypass material development bottlenecks and achieve superior mechanical reliability using currently available materials.

3.3. Case Study: The "Balanced Design" Strategy

We further analyzed specific design cases using a combined PCA projection and SHAP waterfall plot (Fig. 5).

Crucially, the error bars in the waterfall plots represent the standard deviation of SHAP values across the 11 models. The narrow error bars observed for the dominant features (V_{acc}) and (S_{sym}) confirm that the models unanimously agree on their contribution, irrespective of the algorithmic architecture.

- Best Case (Balanced Design): This stack achieves the optimal crease performance. The waterfall plot shows synergistic contributions from all factors: moderate material reduction, high stiffness symmetry ($S_{sym} \approx 0.9$), and controlled elastic energy.

- Second-Best (Extreme Material Optimization): This design relies heavily on extreme thinning of layers and minimization of viscoelasticity but suffers from poor symmetry. While effective, it approaches the physical limits of manufacturing and reliability (e.g., impact resistance).

- Worst Case: Characterized by high viscoelastic accumulation and severe asymmetry, leading to maximum residual deformation.

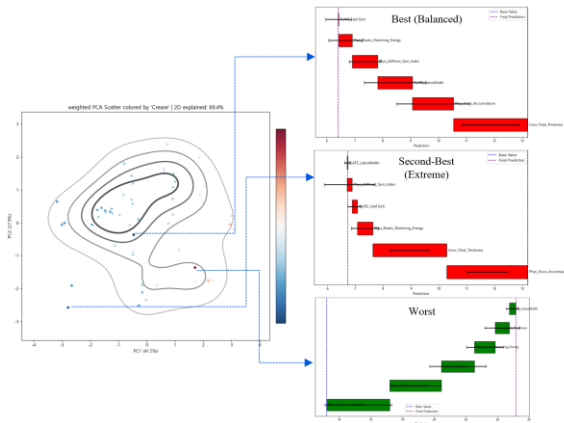


Fig. 5. Comparative Case Study. (Left) PCA projection of the design space showing the clustering of low-crease designs. (Right) SHAP waterfall plots comparing the Best (Balanced) vs. Second-Best (Extreme) designs.

This comparison highlights a crucial engineering insight. While minimizing thickness (as seen in the Second-Best case) is a valid strategy, it often compromises other product attributes. The Best Case demonstrates that structural symmetry can compensate for material limitations. By adopting a "Balanced Design" that prioritizes S_{sym} , engineers can achieve superior crease performance without solely relying on extreme material reduction, resulting in a more robust and practical solution for mass production.

4. Conclusions

This study presents a transformative approach to mechanical design in flexible electronics by establishing a Physics-Informed Agent-Analytics Framework. Moving beyond traditional trial-and-error FEA and opaque black-box ML, we successfully integrated domain knowledge with data science through a rigorous human-AI collaborative workflow. By positioning the

LLM agent as a core "Knowledge Translator," we converted qualitative engineering intuition into high-level mathematical descriptors using the Parallel Axis Theorem (PAT). The framework's validity was rigorously confirmed against a dataset of 50 physically fabricated and experimentally tested foldable stacks, ensuring that our insights reflect real-world phenomenology rather than mere simulation artifacts.

Our comprehensive analysis leads to three pivotal conclusions that redefine the design strategy for next-generation foldable stacks:

Paradigm Shift to "Geometric Compensation": Historically, crease mitigation has relied on material advancements (e.g., improved PSA relaxation). However, our discovery of the Stiffness Symmetry (S_{sym}) factor compels a fundamental shift from "material-dependent" optimization to "geometric compensation." We demonstrated that optimizing the stacking order to neutralize the bending moment can intrinsically reduce the shear load on viscoelastic layers. This proves that structural balance can effectively compensate for material limitations, empowering engineers to achieve superior reliability without waiting for novel material breakthroughs.

"Soft-Elastic" Design Strategy: We challenged the conventional intuition that maximizing stiffness aids recovery. Our analysis of Elastic Strain Energy reveals that excessive stiffness in outer layers accumulates high internal stress, driving plastic deformation in adjacent soft layers. Consequently, we propose a "Soft-Elastic" paradigm—minimizing stored strain energy while maintaining essential rigidity—as a superior strategy for reducing permanent deformation.

Robustness via Balanced Optimization: The comparative analysis of "Balanced" versus "Extreme" designs confirms that a synergistic approach yields the optimal outcome. The "Balanced Design" leverages geometric symmetry to offset material constraints, offering a wider process window and greater mechanical stability than designs relying solely on extreme thickness reduction.

In conclusion, this work demonstrates that a physics-grounded, agent-mediated approach can uncover deep mechanical insights—such as the principle of geometric compensation—that are often missed by purely data-driven methods. This framework paves the way for scalable, AI-accelerated hardware design in the era of complex flexible electronics.

REFERENCES

- [1] Y. Jia, H. Liang, K. Wang, and S. Chen, "Mechanical simulation of foldable AMOLED panel with a module structure," *Organic Electronics*, vol. 65, pp. 185–192, 2019. DOI: 10.1016/j.orgel.2018.11.026.
- [2] D. W. Kim et al., "Fabrication of practical deformable displays: advances and challenges," *Light: Science & Applications*, vol. 12, no. 1, 2023.
- [3] Y. Zhang, S. Wang, F. Dong, et al., "Mechanical Behavior and Constitutive Model Characterization of Optically Clear Adhesive in Flexible Devices," *Micromachines*, vol. 13, no. 2, p. 301, 2022. (DOI: 10.3390/mi13020301).
- [4] P. Xia et al., "Agent0: Unleashing Self-Evolving Agents from Zero Data via Tool-Integrated Reasoning," arXiv preprint arXiv:2511.16043, 2025.